Capstone Project Hardware, Software, Cloud, SAP & API Requirements

Title - Autonomous Industrial IoT Digital Twin & Predictive Maintenance Engine with Closed-Loop ERP Action

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Purpose

This document lists all hardware, software, cloud services, SAP access, API keys, and development resources required for the sponsored SAP Technologies Capstone Project.

Project: Autonomous Industrial IoT Digital Twin & Predictive Maintenance Engine with Closed-Loop ERP Action. The prototype monitors a rotating asset using vibration, temperature, and current sensors; predicts failures using deep learning; and automatically creates SAP PM maintenance actions and SAP MM inventory/procurement actions.

Hardware Requirements

#	Item	Recommended Specification	Qty	Purpose
1	Raspberry Pi Pico W	RP2040, Wi-Fi + Bluetooth, 2MB flash, micro-USB	1	Edge microcontroller for sensor reading and MQTT publishing
2	MPU-6050	6-axis accelerometer + gyroscope, I2C	1–2	Vibration and motion monitoring of motor/bearing
3	DS18B20	Waterproof stainless probe, 1-wire, with 4.7kΩ resistor	1	Bearing/motor temperature monitoring
4	ACS712	5A or 20A Hall-effect current sensor	1	Motor current/load monitoring
5	DC motor	5V–12V DC motor with shaft	1	Test rotating asset
7	Motor driver	L298N or MOSFET driver module	1	Controls motor speed and load
8	Power supply	5V 2A USB adapter + 12V 1–2A adapter	1 each	Pico and motor power
9	Breadboard	830-point solderless breadboard	1–2	Prototyping
10	Jumper wires	Male-female and male-male, 20–30 each	1 set	Connections

11	Resistors/capacitors	4.7kΩ, 10kΩ, 100nF, 10µF assortment	1 kit	Signal conditioning and stabilization
12	Micro-USB cable	Data-capable cable	1	Programming and power
13	SD card module + microSD	SPI microSD module + 8–16GB card	1	Local buffering if Wi-Fi fails
14	OLED display	0.96-inch I2C SSD1306	1	Local edge status display
15	Enclosure / mounting	vibration-damping pads	1	Secure sensor mounting

Cloud Requirements — AWS

AWS Account and Services

AWS Service	Required Access	Purpose
AWS Account	Sponsored/root account or IAM access	Cloud hosting and IoT pipeline
AWS IoT Core	Thing creation, certificates, IoT Rules	MQTT ingestion and routing
AWS Lambda	Function creation and execution	Serverless processing and inference
Amazon S3	Bucket creation/read/write	Raw data, datasets, model artifacts
Amazon Timestream	Database/table creation	Time-series telemetry
Amazon DynamoDB	Table creation/read/write	Asset metadata, alerts, event state
Amazon SNS	Topic creation and subscriptions	Email/SMS alerts
Amazon CloudWatch	Logs, metrics, alarms	Monitoring and debugging
IAM	Roles, policies, users	Secure access control
AWS Secrets Manager	Secret storage	Secure storage of keys/credentials

AWS Credentials / Access Required

1. AWS account ID and recommended region
2. IAM user or SSO access with permissions for AWS IoT Core, Lambda, S3, Timestream, DynamoDB, SNS, CloudWatch, IAM (limited), and Secrets Manager
3. AWS access key ID + secret access key (development use only)
4. AWS IoT Core endpoint URL
5. Device certificate, private key, and Amazon Root CA
6. S3 bucket name and access policy
7. Timestream database/table name
8. DynamoDB table names
9. SNS topic ARN and subscriber email
10. Budget/billing alerts configuration

AWS IoT Core's 12-month Free Tier includes 2.25 million connection minutes, 500,000 messages, 225,000 Registry/Device Shadow operations, and 250,000 rule triggers/actions per month; AWS Lambda's free tier includes one million requests and 400,000 GB-seconds per month.

SAP Requirements

SAP Access Required

SAP Resource	Required Access	Purpose
SAP BTP Trial/Free-Tier account	Subaccount, space, entitlements	Integration and extension platform
SAP Integration Suite	Integration flow design/deploy access	AWS-to-SAP API orchestration
SAP API Management	API provider/proxy access	Secure API exposure
SAP Destination Service	Destination configuration	Secure S/4HANA connectivity
SAP S/4HANA Sandbox	Business user + technical user	PM and MM business processes
SAP S/4HANA OData APIs	API access/communication arrangement	Create/read maintenance notifications and orders
SAP PM authorization	Notification/order creation	Maintenance workflow automation
SAP MM authorization	Material/stock read, PR creation	Spare-part and procurement workflow
SAP Fiori Launchpad	Developer/user access	Dashboard and app deployment
SAP Build Apps / SAPUI5	App development access	Fiori-like dashboard development
SAP AI Core / Generative AI hub	Optional model/LLM access	Advanced AI integration

SAP Credentials Required

1. SAP BTP trial or sponsored subaccount URL
2. BTP subdomain and region
3. BTP user ID and password / SSO access
4. Entitlements for SAP Integration Suite, API Management, Destination Service, Build Apps/UI5, and optional AI Core
5. SAP S/4HANA sandbox URL and client number
6. Business user ID and password
7. Technical/communication user ID for OData APIs
8. OAuth client ID and client secret
9. OData API endpoint URLs
10. Plant code, maintenance plant, planner group, work center
11. Equipment number and functional location for the demo asset
12. Material numbers for spare parts
13. Storage location and purchasing organization/group
14. SAP API key / communication arrangement details

AI / LLM API Requirements

API / Service	Required Credential	Purpose
OpenAI API	API key + organization ID	LLM reasoning and RAG-based diagnosis
Anthropic API	API key	Alternative LLM for agents
Google Gemini API	API key	Alternative LLM
LangSmith / LangFuse	API key	Agent tracing and observability